

Automotive Vision Capture Plugin for Instrument Studio (PXIe-148X)

This plugin provides an easy-to-use **automotive camera (FPD-Link/ GMSL) acquisition, display, logging** workflow in **Instrument Studio 2025**, powered by NI FlexRIO **PXIe-148X** modules (e.g., **PXIe-1486**).

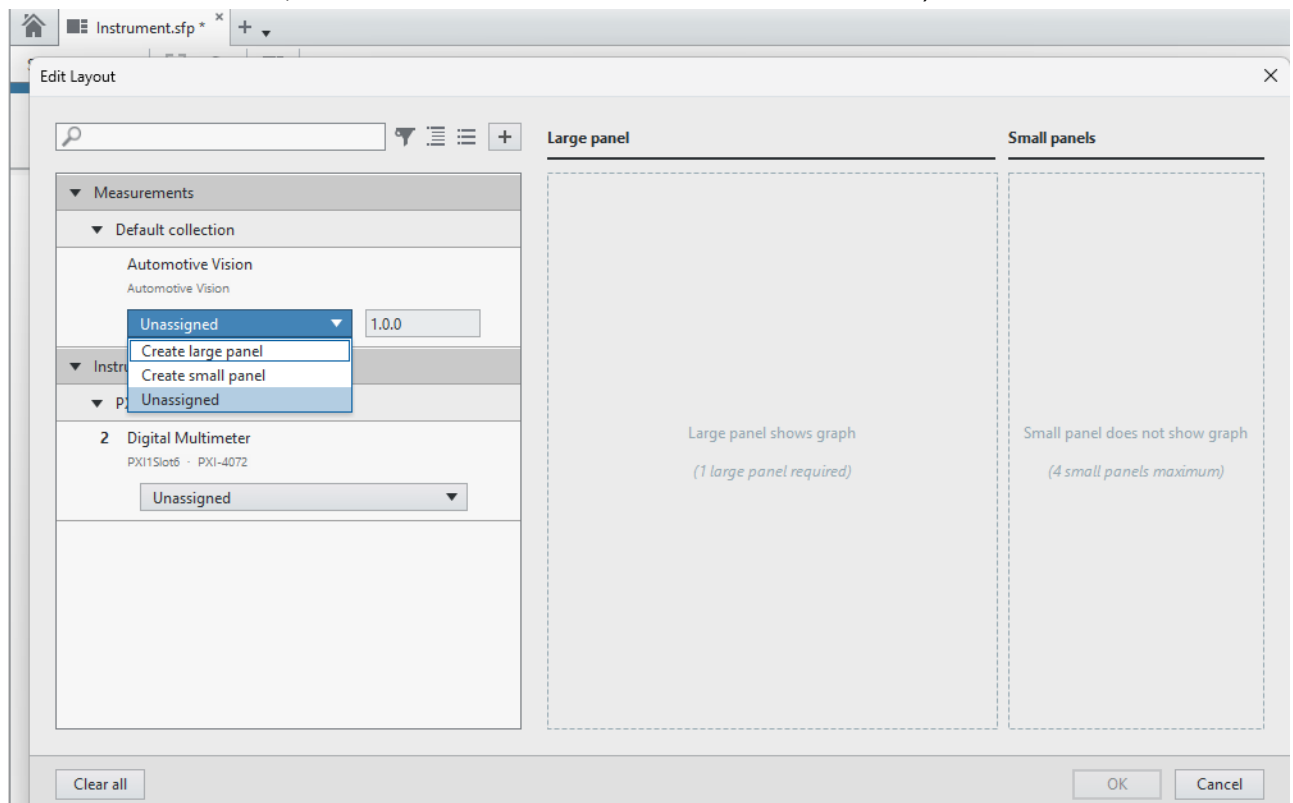
Prerequisites

- **LabVIEW 2025 Q3** with:
 - **NI-FlexRIO** driver (23Q1 or newer) and **PXIe-148X** example set.
 - **Vision Development Module** (IMAQ/IMAQdx for image handling & decoding).
- **Instrument Studio 2025 Q4**.
- Supported **PXIe-148X** module (e.g., **PXIe-1486 8-In or 4-In/4-Out**) with camera and FAKRA coax cabling; PoC capable power as required.

How to Build & Run the Plugin

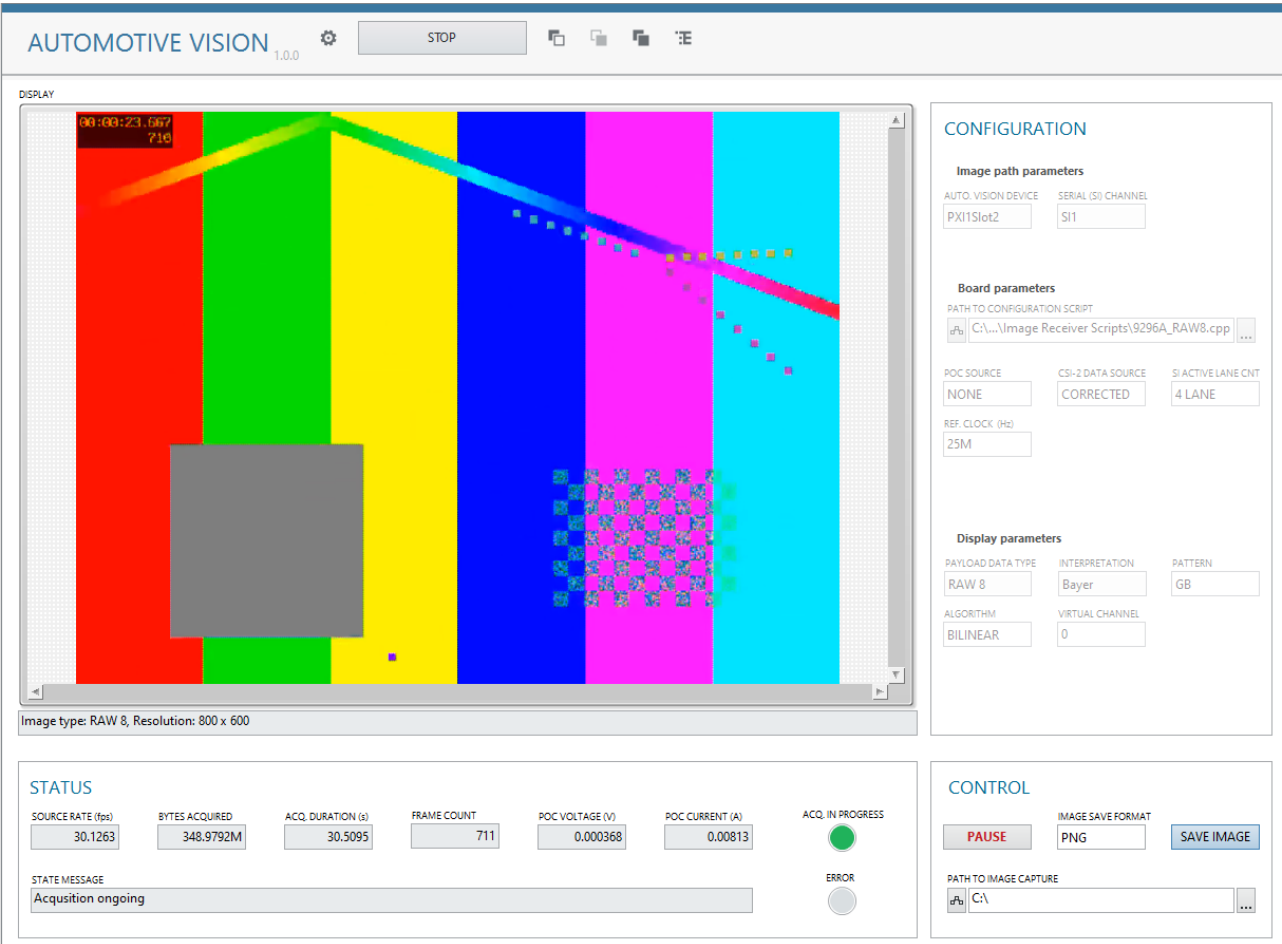
1. Open Instrument Studio (IS) 2025

Create a **Manual Layout**, add a **Large Panel** and select **Automotive Vision Capture** (pay attention to the monitor resolution; a minimum 1920 x 1080 resolution should be used).



2. UI in IS

The plugin panel appears with live image, packet/trace tabs and status telemetry.



Configuration

These controls map to NI's PXIe-148X acquisition example front panel. Use values matching the **camera configuration script**.

CONFIGURATION

Image path parameters

AUTO. VISION DEVICE

PXI1Slot2

SERIAL (SI) CHANNEL

SI1

Board parameters

PATH TO CONFIGURATION SCRIPT

C:\...\Image Receiver Scripts\9296A_RAW8.cpp

...

POC SOURCE

NONE

CSI-2 DATA SOURCE

CORRECTED

SI ACTIVE LANE CNT

4 LANE

REF. CLOCK (Hz)

25M

Display parameters

PAYLOAD DATA TYPE

RAW 8

INTERPRETATION

Bayer

PATTERN

GB

ALGORITHM

BILINEAR

VIRTUAL CHANNEL

0

CSI-2 Data Source

Select **RAW** or **Corrected** payload stream as provided by the camera/deserializer path.

TIP: RAW = sensor output; Corrected = post-processed/error-corrected.

Active Lane Count

Choose **1 / 2 / 4 lanes** for CSI-2 throughput. Higher lane count → higher bandwidth for high-res/FPS.
(PXIe-1486 supports 4-lane CSI-2, 800 Mbps/lane.)

Reference Clock Source

Set the **refclk** used for link sync (typically **25 MHz**, camera-dependent; input channels commonly **23–26 MHz**).
Ensure it matches camera PLL settings.

Payload Data Type

Select pixel format (**RAW8**, **YUV422**, **RGB888**, etc.), exactly as configured in the **camera's script**, so the decode/display matches the stream.

Interpretation

Choose how payload is decoded for display: **grayscale**, **Bayer**, **RGB**, **YUV420/422**. This must match the **data type** and the **camera output**.

Pattern (Bayer only)

If **Bayer** is selected, set the CFA pattern: **RG / BG / GR / GB** to reconstruct correct colors.

Algorithm (Bayer demosaicing)

Pick **Bilinear** (fast) or **VNG** (higher edge quality, slower). NI recommends Bilinear first; use VNG where edge fidelity matters.

Virtual Channel

Select the **CSI-2 Virtual Channel ID** to demultiplex multiple streams sharing the link. Useful for multi-sensor aggregates.

PoC (Power-over-Coax)

- Choose **PoC source**, observe **PoC voltage/current** indicators for camera power health.

Configuration — ** Board **

RIO Device & Bitfile

- Bitfile selected automatically based on your hardware selection.

AUTO VISION DEVICE

- Select which device to use.

SERIAL (SI) CHANNEL

- Select which serial input channel to use - only one channel is supported at a time.

Starting & Using the Plugin

1. **RUN** the plugin in Instrument Studio.

It connects to the selected serial channel(s), starts acquiring CSI-2 packets and image frames; status LED shows connection. The service updates the UI periodically (based on FPS). On higher resolution the FPS can drop.

2. **View live image & packets**

- **Image pane** shows decoded frames.

3. **Interact via controls**

- **Pause / Resume** (OOB gRPC) to suspend UI updates without stopping acquisition.

- **Capture:** save one frame correspond to the settings above.
- **Capture type:** can be PNG or RAW.
- **Capture path:** a folder to save the capture - .bin or .png file.

The CONTROL panel includes a **PAUSE** button, an **IMAGE SAVE FORMAT** dropdown menu set to **PNG**, and a **SAVE IMAGE** button. Below these is a **PATH TO IMAGE CAPTURE** text field with a file explorer icon on the left and a dropdown arrow on the right, currently showing **C:**.

Indicators (Status Telemetry)

The STATUS panel displays several indicators: **SOURCE RATE (fps)** at 0 FPS, **BYTES ACQUIRED** at 0, **ACQ. DURATION (s)** at 0s, **FRAME COUNT** at 0, **POC VOLTAGE (V)** at 0V, **POC CURRENT (mA)** at 0A, and **ACQ. IN PROGRESS** (indicated by a grey circle). Below these is a **STATE MESSAGE** text area and an **ERROR** indicator (indicated by a grey circle).

- **Source rate (FPS)** – current frame rate per channel.
- **Bytes acquired** – total data captured this session.
- **Acquisition duration (s)** – elapsed time.
- **Frame count** – total frames received.
- **PoC voltage / current** – camera power telemetry over coax.
- **Acquisition in progress** – Indicates that the AutoVision device is capable of receiving frames.
- **Error** – Indicates error in measurement logic.

Tips & Known Behaviors

- **Match scripts to UI:** Payload type, interpretation, Bayer pattern and VC must align with the **camera configuration script**, else display/packet parsing will be wrong.
 - **Throughput:** CSI-2 lane count and refclk affect bandwidth. PXIe-1486 supports **4-lane CSI-2 @ ~800 Mbps/lane** per NI docs.
 - **Display channel count:** Default host/FPGA code displays up to **one** channels; more requires code edits.
 - **Displaying performance:** Displaying **very large resolution or high FPS** may cause the VI to appear unresponsive while processing. Displayed FPS are reduced in these case.
 - **PoC safety:** Monitor **PoC V/I**; ensure proper termination and power budgets for the sensor.
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